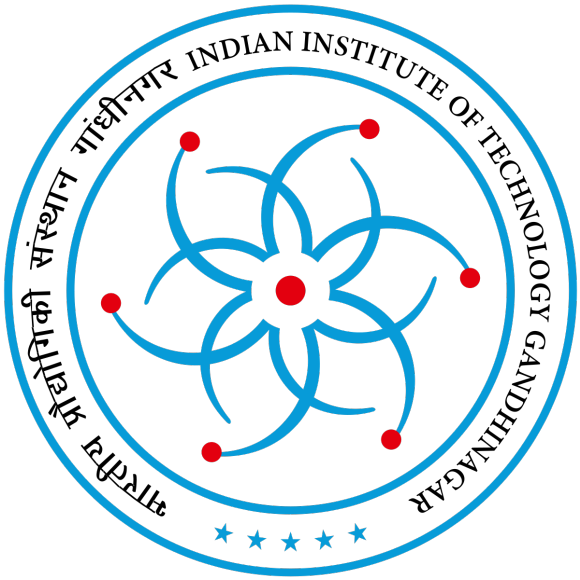
Indian Institute of Technology Gandhinagar



Velocity and Acceleration Analysis of a Four-Bar Mechanism for Upper-Arm Rehabilitation

ME 335 Synthesis and Analysis of Mechanisms

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1 Introduction

In the first phase of this project, a four-bar crank–rocker mechanism was synthesized for upper-arm rehabilitation based on experimentally recorded human arm motion. The earlier phase focused on mechanism type selection, collection and processing of motion capture data, extraction of representative arm configurations, and the graphical synthesis of a feasible linkage capable of approximating the desired rehabilitation motion. The resulting design was verified at a conceptual level through kinematic feasibility checks and software-based motion simulation.

The present second phase builds directly on that synthesized mechanism and shifts the focus from *mechanism synthesis* to *mechanism analysis*. Having obtained a feasible mechanism in Phase 1, the objective now is to study its motion in detail through kinematic analysis. This includes motion depiction using both a CAD-based model and a fabricated prototype, followed by position, velocity, and acceleration analysis of the mechanism. Through this phase, the motion characteristics of the rehabilitation linkage are examined quantitatively to understand how the input motion is transmitted through the links and how closely the generated output matches the intended upper-arm rehabilitation trajectory.

2 Objective

The objective of Phase 2 is to analyze the kinematic performance of the synthesized four-bar rehabilitation mechanism. In particular, the aim is to validate the motion of the mechanism using MSC ADAMS and a fabricated prototype, and to study how a constant-speed rotary input is transformed into the desired oscillatory upper-arm motion through the position, velocity, and acceleration characteristics of the main moving links.

3 Mechanism Description & Continuity from Phase 1

The mechanism analyzed in this phase is based on the crank–rocker four-bar linkage synthesized in Phase 1 for upper-arm rehabilitation. While retaining the overall linkage configuration, minor geometric modifications were introduced to improve motion characteristics. Specifically, the crank length was reduced, and the distance between the ground pivots was slightly adjusted to ensure that the mechanism continues to pass through the same target points as before. These changes were aimed at reducing jerk and achieving smoother motion. The present work focuses on evaluating the kinematic performance of the modified mechanism rather than redesigning it entirely.

From the rehabilitation perspective, a continuously rotating input crank is advantageous because it can be driven by a motor at controlled speed, while the rocker output naturally produces a bounded oscillatory motion suitable for guided arm movement. Therefore, the main question addressed in this phase is whether the synthesized mechanism produces smooth, repeatable, and kinematically acceptable motion when subjected to a prescribed actuation input.

4 ADAMS Model Development

The synthesized four-bar mechanism obtained in Phase 1 was modeled and analyzed using MSC ADAMS to study its kinematic behavior under realistic motion conditions. The geometry of each link was recreated in ADAMS, and appropriate joints were assigned to replicate the planar mechanism.

Revolute joints were defined at all pin connections, and the ground link was fixed to establish a reference frame. A rotary motion input was applied to the crank link to simulate continuous actuation, consistent with the crank–rocker configuration identified during synthesis.

The mechanism was simulated over a full cycle of motion to verify smooth operation, absence of link interference, and proper transmission of motion from input to output. The ADAMS environment also enabled extraction of position, velocity, and acceleration data for further kinematic analysis.

This modeling step ensures that the synthesized mechanism is not only geometrically valid but also kinematically consistent when subjected to realistic actuation conditions.

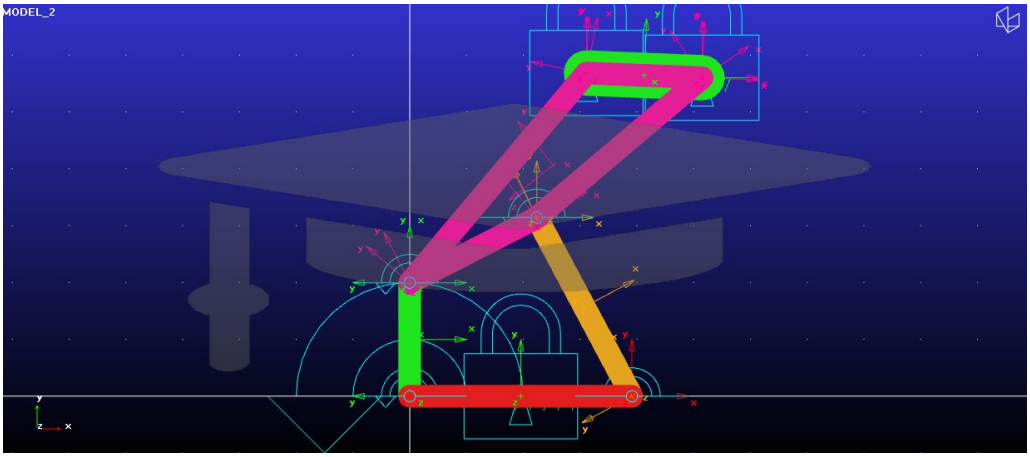


Figure 1: ADAMS model of the synthesized four-bar crank–rocker mechanism showing link configuration, joints, and motion setup.

4.1 Procedure of Making Four Bar Mechanism in ADAMS

1. A new model was created in MSC Adams and appropriate unit settings were defined.
2. The ground link was fixed to serve as the reference frame for the mechanism.
3. Rigid bodies representing the crank, coupler, and rocker were created.
4. The links were positioned according to the required geometric dimensions (lengths and initial angles).
5. The links were connected using revolute joints to form a closed kinematic chain.
6. A rotational motion input was applied to the crank link to drive the mechanism.
7. The simulation was executed for a specified time duration.

8. Post-processing tools were used to extract the velocity and acceleration of selected points.

9. The obtained plots were analyzed to study:

    - Variation of motion parameters with time
    - Key trends in the response
    - Peak values of velocity and acceleration

5 CAD Model Assembly

The figure below shows the CAD Model for the four bar mechanism which was designed in Autodesk Fusion 360. The link Lengths were updated for an optimized mechanism. All the dimensions can be found in CAD Drawings whose link can be found attached at the end of the report.

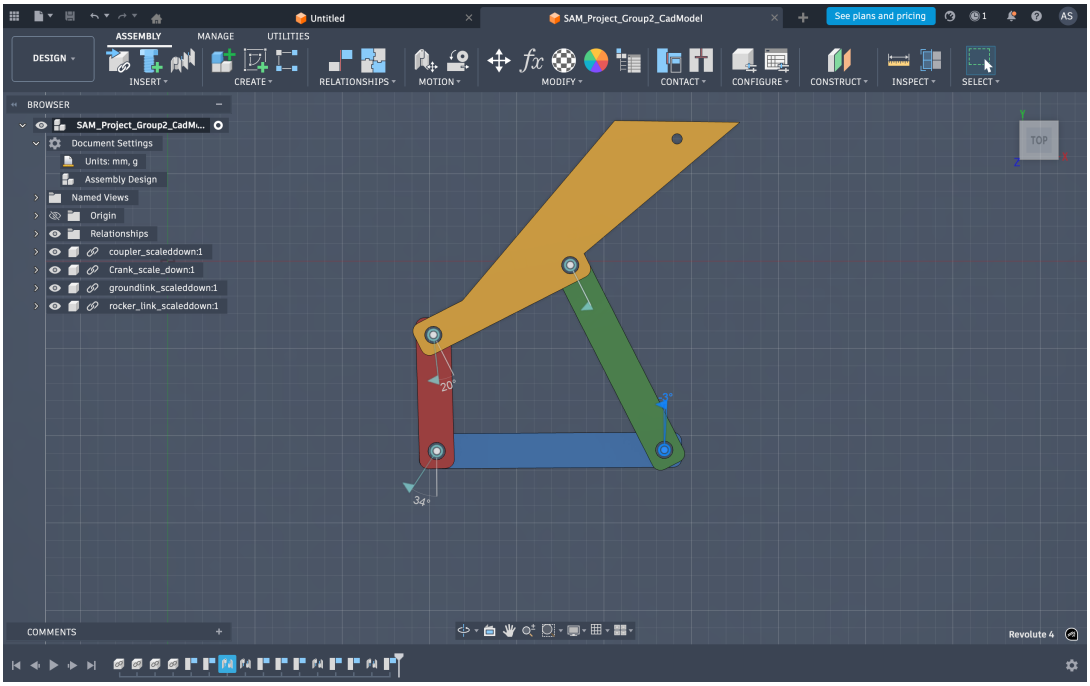


Figure 2: CAD Model

- The Blue Link is the Ground Link.
- The Red Link is the Crank.
- The Green Link is the Rocker.
- The Yellow Link is the Coupler.

6 Physical Model and Motion Depiction

A physical prototype of the synthesized four-bar crank–rocker mechanism was fabricated using laser-cut links and pin joints. The model was mounted on a rigid base to represent the ground link, and all joints were implemented using simple revolute connections. Please Note that the same color coding has been used for the CAD Model and the Physical Prototype. The Light Blue Link in the image is the forearm of the patient and the Orange Link is the Upper Arm.

The mechanism was manually actuated to demonstrate the motion of the rehabilitation system. The observed motion confirmed smooth transmission from the input crank to the output rocker, closely matching the intended upper-arm movement obtained during the synthesis phase.

This physical realization serves as a validation of the design and provides a tangible representation of the mechanism's functionality.

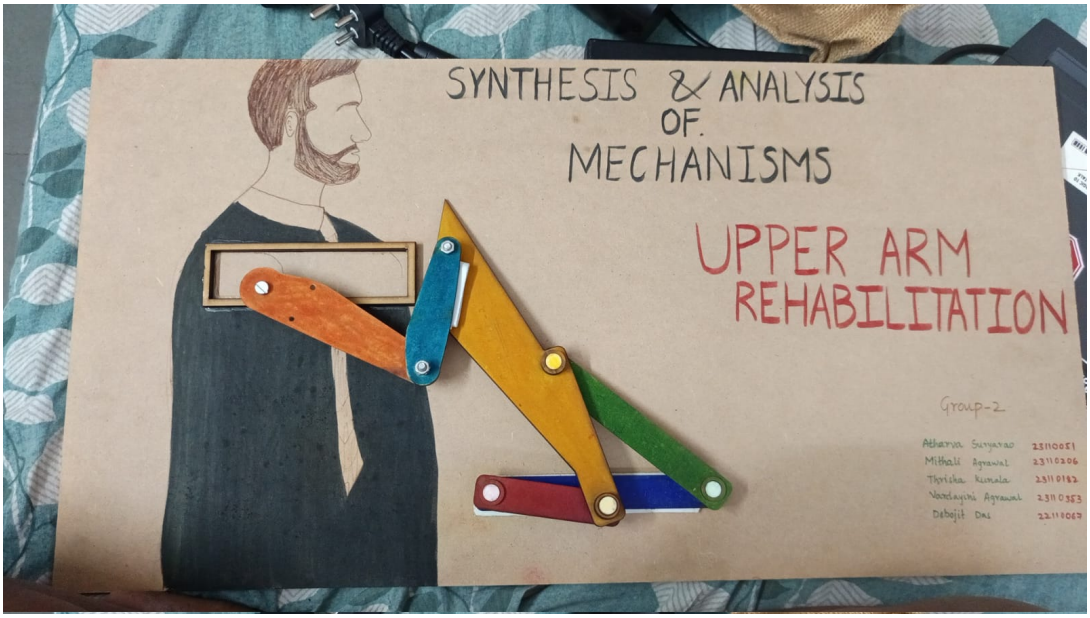


Figure 3: Fabricated physical model of the four-bar rehabilitation mechanism demonstrating motion using laser-cut links and revolute joints.

7 ADAMS-Based Kinematic Analysis

The synthesized rehabilitation mechanism was further analyzed in MSC ADAMS by prescribing a constant angular velocity input to the driving revolute joint. The input motion was defined as a rotational velocity function of magnitude 1.047 rad/s, which corresponds approximately to 60°/s. This provides a controlled and repeatable basis for studying the kinematic response of the remaining links.

A meaningful interpretation of the ADAMS plots requires distinguishing between the input link and the driven links. The input link is expected to exhibit nearly constant angular velocity and almost zero angular acceleration because its motion is prescribed directly. In contrast, the forearm and rocker links are not directly driven and hence their

motion is generated by the geometry of the four-bar mechanism. Hence, their angular and linear kinematic quantities vary with time and reflect the motion-transmission characteristics of the linkage.

Overall, the ADAMS results confirm the expected behavior of a crank–rocker mechanism. The input rotation remains essentially uniform, whereas the output and coupler-associated links undergo non-uniform oscillatory motion. This is desirable in the present application because the rehabilitation link is not intended to rotate continuously, but instead to move through a finite therapeutic range. At the same time, the plots show that the motion is continuous over the simulated interval, indicating that the synthesized mechanism is kinematically feasible.

7.1 Input Motion Definition

The imposed input motion is shown in Fig. 4. A constant angular velocity was specified at the input revolute joint. This input serves as the reference against which the motion of the remaining links is interpreted.

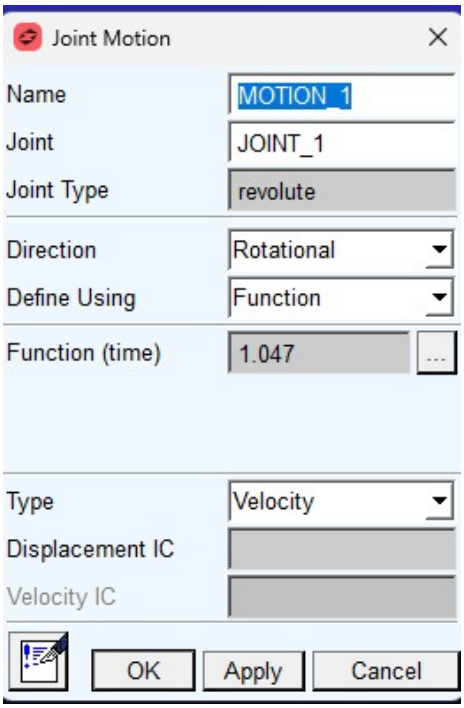


Figure 4: Input motion definition used in ADAMS. A constant rotational velocity of 1.047 rad/s was applied which corresponds to 10 revolutions per minute at the input revolute joint.

7.2 Angular Velocity & Acceleration of Crank Link

The first set of ADAMS plots, Figs. 5 shows that (Crank) has nearly constant angular velocity and almost zero angular acceleration throughout the simulation interval. This is consistent with the assigned ADAMS input and indicates that (Crank) behaves as the driving crank. This agreement is important because it validates the simulation setup before interpreting the remaining plots.

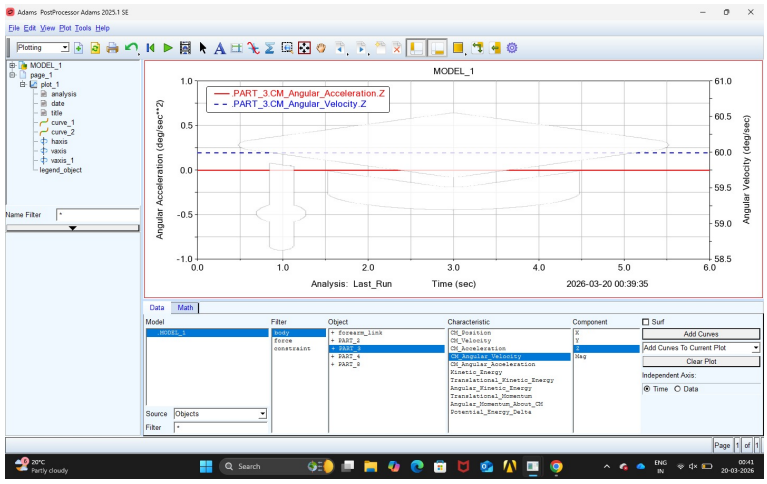


Figure 5: Angular kinematic response from ADAMS. The first plot shows the angular velocity and angular acceleration of the Center of Mass of PART_3 (Crank).

7.3 Angular Velocity & Acceleration of Rocker Link

The second set of ADAMS plots establishes the distinction between the input and output responses. In Fig. 5, (Rocker) shows time-varying angular velocity and angular acceleration. Its angular velocity changes sign during the cycle, confirming oscillatory motion rather than continuous rotation. This is the expected behavior of the rocker in a crank–rocker mechanism.

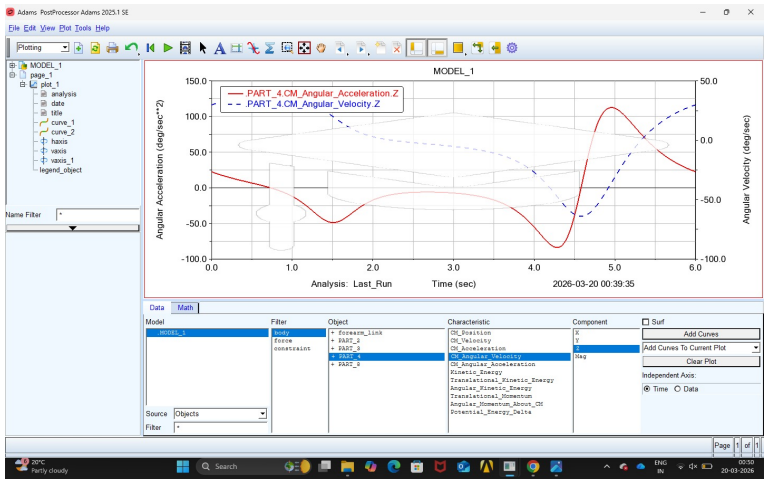


Figure 6: Angular kinematic response from ADAMS. The plot shows the angular velocity and angular acceleration of the Center of Mass of PART_4 (Rocker).

7.4 Linear Velocity and Acceleration of Crank

The linear kinematic plots of (Crank) are shown in Fig.6 and Fig.7 . Since (Crank) acts as the driving link, its motion remains more regular than that of the forearm and rocker links. The component variations are smooth and periodic, and the velocity magnitude remains bounded with no abrupt discontinuity. This indicates that the input link motion is being transmitted in a stable way through the mechanism.

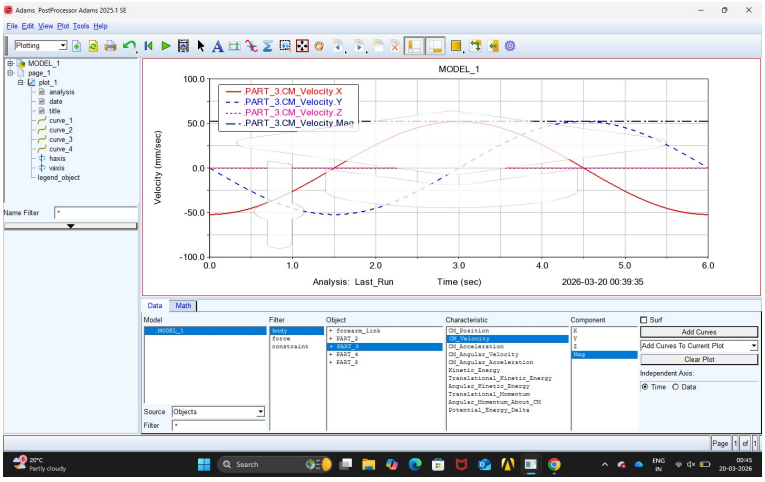


Figure 7: Linear velocity of center of mass of the Crank

The acceleration components of (Crank) also remain relatively moderate compared to the stronger peaks seen in the forearm link. This is expected because the input crank has a prescribed angular speed and therefore does not undergo the same reversal-type behavior as the output link. These plots reinforce the interpretation that the main non-uniformity of the motion arises from the geometry of the mechanism rather than from instability in the input.

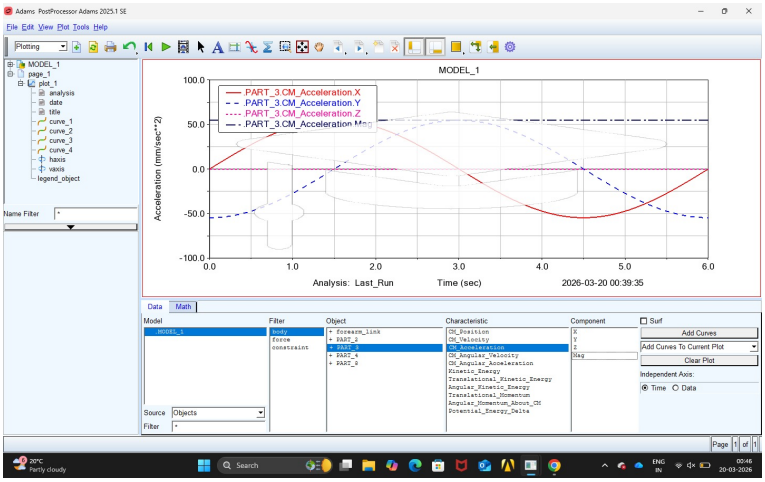


Figure 8: Linear acceleration of center of mass of the Crank

7.5 Linear Velocity and Acceleration of Rocker Link

The linear velocity and acceleration of (Rocker) are shown in Fig.8. Compared to (Crank), (Rocker) shows stronger variation in both component values and overall magnitude. This is consistent with its role as an oscillating output link. As the rocker approaches its extreme positions and reverses direction, the corresponding velocity and acceleration components change more sharply.

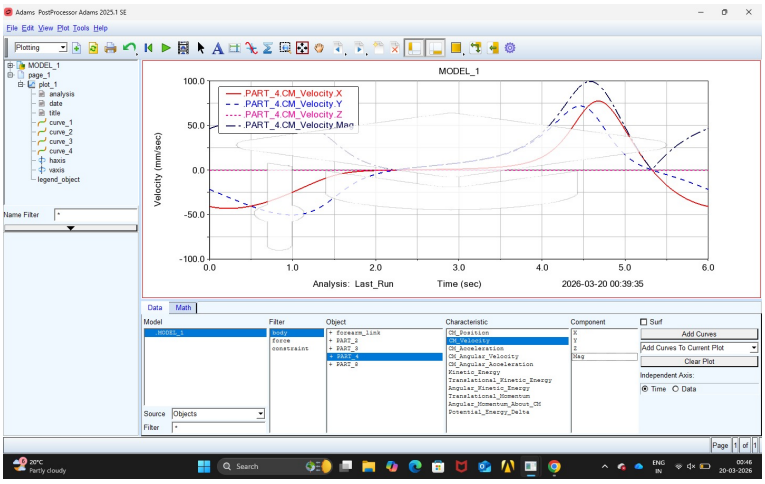


Figure 9: Linear velocity of center of mass of the Rocker Link.

The plots therefore provide a direct kinematic signature of output oscillation. The velocity changes sign over the cycle, which confirms reversal of direction, while the acceleration peaks occur near these reversal regions. Such behavior is expected in four-bar mechanisms and is not a numerical anomaly. Instead, it indicates that the mechanism is functioning according to the crank–rocker principle established during synthesis.

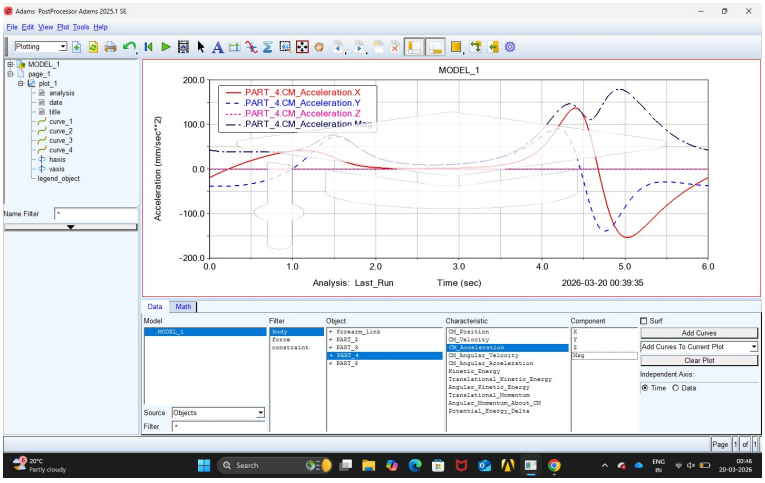


Figure 10: Linear acceleration of center of mass of the Rocker Link

7.6 Angular Motion of the Forearm Link

The forearm link is the most critical component from the rehabilitation perspective, as it directly governs the guided motion of the patient's arm. The angular velocity and angular acceleration plots (Fig. 10) reveal the following key characteristics:

- **Non-Uniform but Continuous Motion:** The angular velocity varies non-uniformly throughout the cycle, while remaining smooth and continuous. This confirms that a constant-speed input crank is transformed into a variable output motion.
- **Acceleration Peaks at Transition Zones:** The angular acceleration exhibits significant positive and negative peaks, particularly near the reversal positions of

the rocker. These peaks arise due to rapid changes in the transmission angle of the mechanism.

- **Transmission Angle Influence:** These acceleration peaks are associated with variations in the transmission angle of the mechanism. As the transmission angle deviates from its optimal range, the efficiency of motion transfer reduces, leading to higher acceleration fluctuations.
- **Effect of Four-Bar Geometry:** The observed variations are inherent to crank–rocker mechanisms and reflect nonlinear motion transmission rather than any instability in the system.
- **Rehabilitation Significance:** The non-uniform motion is beneficial because:
  - Slower motion phases allow controlled muscle engagement
  - Faster return phases reduce fatigue and improve efficiency
- **Smoothness and Jerk Consideration:** The continuity of the acceleration curve indicates finite jerk, ensuring:
  - No sudden shocks or impulses
  - Comfortable and safe interaction for the patient

As seen in Fig. 10, the sharp peaks in angular acceleration occur near the reversal positions of the rocker, where the direction of motion changes. These regions represent critical transition points in the rehabilitation cycle. Overall, the angular motion demonstrates that the mechanism produces a smooth, controlled, and physiologically relevant rehabilitation trajectory.

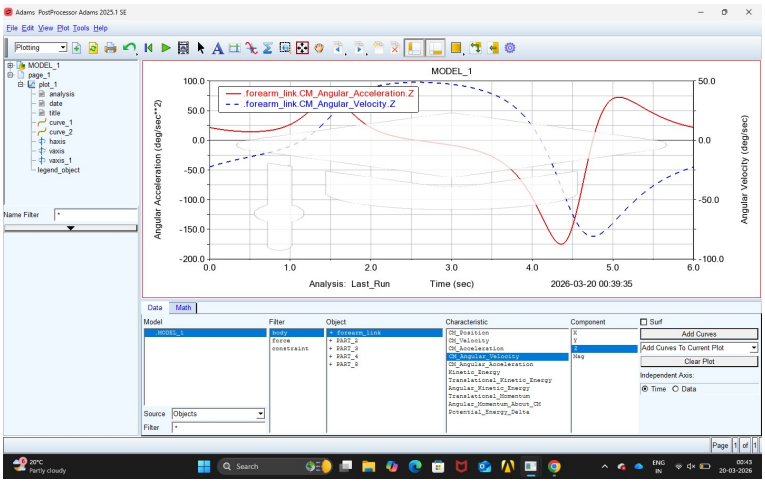


Figure 11: Angular acceleration and angular velocity of the forearm link

7.7 Linear Velocity and Acceleration of the Forearm Link

The linear velocity and linear acceleration of the forearm link center of mass are shown in Fig.11 and Fig.12. These plots provide additional evidence that the motion is predominantly planar. The x- and y-components vary significantly with time, whereas the

z-component remains close to zero throughout the motion. This is consistent with the planar synthesis assumptions made in Phase 1 and confirms that the ADAMS model does not exhibit unwanted out-of-plane motion.

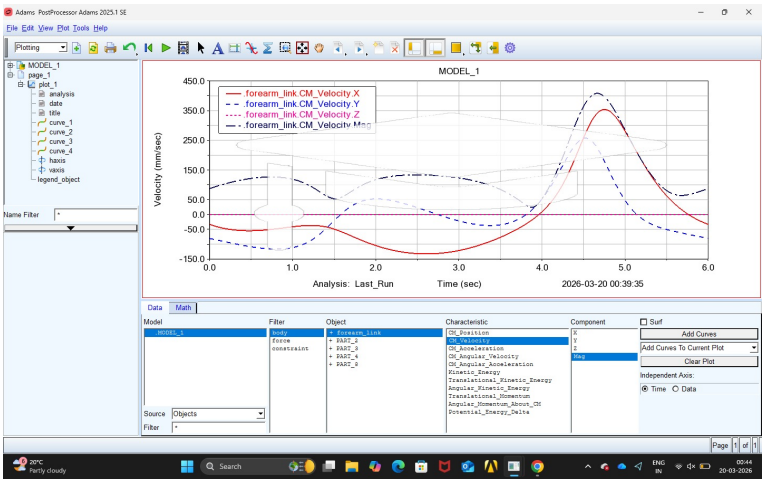


Figure 12: Linear velocity of center of mass of the forearm link.

The magnitude curves show that the linear speed and acceleration increase substantially in certain portions of the motion cycle. These peaks correspond to intervals where the angular motion of the forearm also changes rapidly. Therefore, the translational and rotational plots support each other and together describe the same physical phenomenon: a uniform crank input is being transformed into non-uniform rehabilitation motion.

The maximum linear velocity of the forearm link is observed to be approximately 0.35 m/s, while the corresponding acceleration reaches peak values of the order of 1–1.5 m/s^2. These values fall within acceptable limits for upper-limb rehabilitation devices, ensuring safe and controlled motion.

These acceleration peaks correspond to increased inertial forces and therefore higher torque demand on the actuator. This highlights the importance of selecting a motor capable of handling peak load conditions without compromising smooth operation.

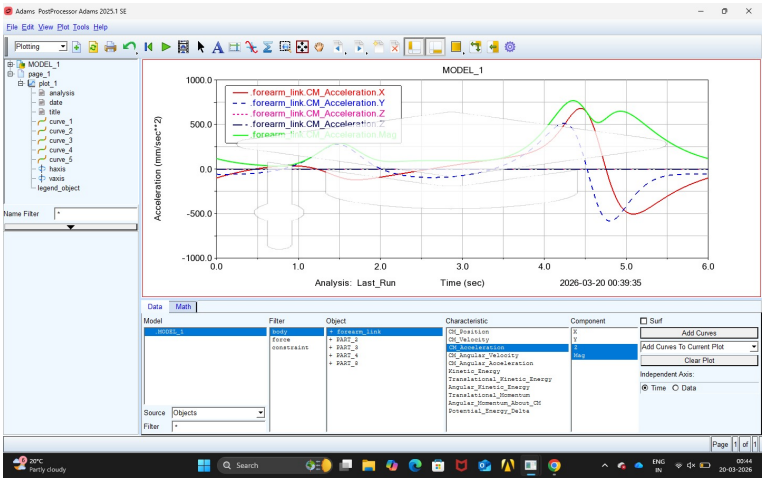


Figure 13: Linear acceleration of center of mass of the forearm link

7.8 Summary of the ADAMS Results

Taken together, the ADAMS plots provide a coherent picture of the synthesized mechanism. The input link rotates with nearly constant angular velocity, verifying correct motion prescription. The forearm and output links respond with non-uniform angular and linear motion, which is a natural consequence of four-bar linkage geometry. The motion remains planar and continuous, indicating that the model is kinematically well-posed.

A key observation is the appearance of relatively large acceleration peaks in the forearm and output links. These occur near reversal positions and rapid transmission-change regions. Such peaks are common in crank–rocker mechanisms, but from the rehabilitation viewpoint they are important because excessive acceleration may produce jerky motion or reduce patient comfort. Therefore, the present mechanism can be considered kinematically successful, while also motivating future refinement of link proportions or input-speed scheduling to reduce acceleration spikes.

8 Comparison of ADAMS and Exported Kinematic Data From Motiongen

In addition to the direct post-processing plots obtained from MSC ADAMS, the kinematic data from MotionGen shared in spreadsheet form for the crank arm, forearm link, rocker arm, and the link table were also plotted against time. These comparison plots provide a cleaner consolidated view of the angular displacement, angular velocity, and angular acceleration of the key moving members on a common time axis. Thus, they complement the earlier ADAMS screenshots and make the motion trends easier to interpret.

The comparison plots are consistent with the earlier ADAMS-based observations. The crank arm exhibits nearly constant angular velocity and essentially zero angular acceleration, confirming that it acts as the prescribed input link. Its angular displacement increases almost linearly with time, with the jump at the end corresponding to angular wrap-around from 2π back to 0, rather than any physical discontinuity in motion.

The forearm and rocker links show the expected non-uniform response of a crank–rocker mechanism. Their angular displacement curves remain bounded, indicating oscillatory rather than continuously rotating motion. The angular velocity of these links changes significantly over the cycle, and the sign change confirms reversal of motion near the extreme positions. This agrees with the previous ADAMS plots, where the output-related links showed time-varying angular velocity and acceleration.

The angular acceleration comparison is particularly informative. The crank arm remains close to zero acceleration throughout, while the forearm and rocker exhibit alternating positive and negative peaks. These peaks occur near the reversal regions, where the mechanism undergoes rapid change in motion transmission. This observation directly supports the earlier interpretation from the ADAMS plots that the mechanism is kinematically feasible, but that some configurations produce sharper acceleration changes which may require further smoothing for rehabilitation use.

The rocker arm and link table curves are very close to each other in both velocity and acceleration. This indicates that the link table follows the output motion very closely,

which is consistent with its role as the arm-supporting member attached to the mechanism. Such agreement is useful because it shows that the motion delivered to the rehabilitation interface is consistent with the expected rocker motion.

Overall, these consolidated plots strengthen the conclusions drawn from the original ADAMS post-processor figures. The previous plots helped identify the motion characteristics of individual parts in detail, while the present combined plots make the relative behavior of all main links visually clear in a compact form. Together, both sets of results confirm that the synthesized mechanism behaves as a crank–rocker with a constant-speed input and a bounded oscillatory rehabilitation output.

8.1 Angular Displacement Comparison

Figure 14 compares the angular displacement of the crank arm, forearm link, rocker arm, and link table over the full motion cycle. The crank arm shows an almost linear displacement increase, consistent with constant-speed input rotation. The forearm and rocker links remain within a limited angular range, confirming their oscillatory response.

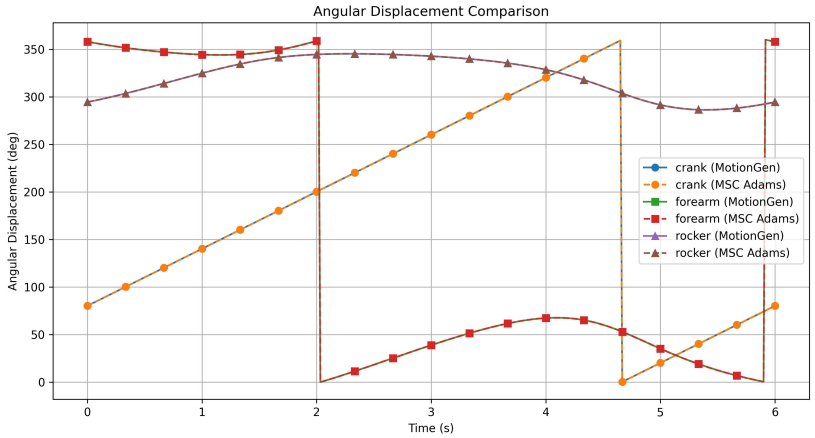


Figure 14: Comparison of angular displacement versus time for the crank arm, forearm link and the rocker arm.

8.2 Angular Velocity Comparison

Figure 15 compares the angular velocity histories of the major links. The crank arm remains constant throughout the cycle, while the forearm and rocker velocities vary strongly with time. The sign change in the forearm and rocker curves indicates reversal of direction at their extreme positions.

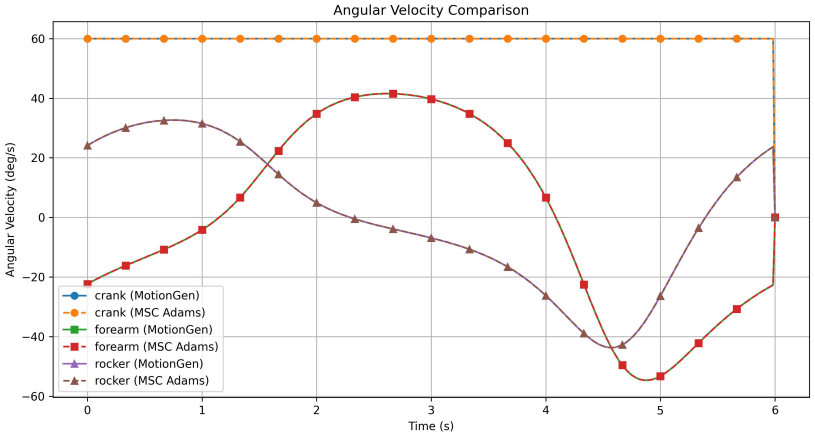


Figure 15: Comparison of angular velocity versus time for the crank arm, forearm link and the rocker arm

8.3 Angular Acceleration Comparison

Figure 16 shows the angular acceleration response of the same members. The crank arm remains close to zero acceleration, as expected for a constant-speed input. The forearm and rocker links exhibit acceleration peaks near reversal regions, which is typical of four-bar motion transmission and is consistent with the earlier ADAMS plots.

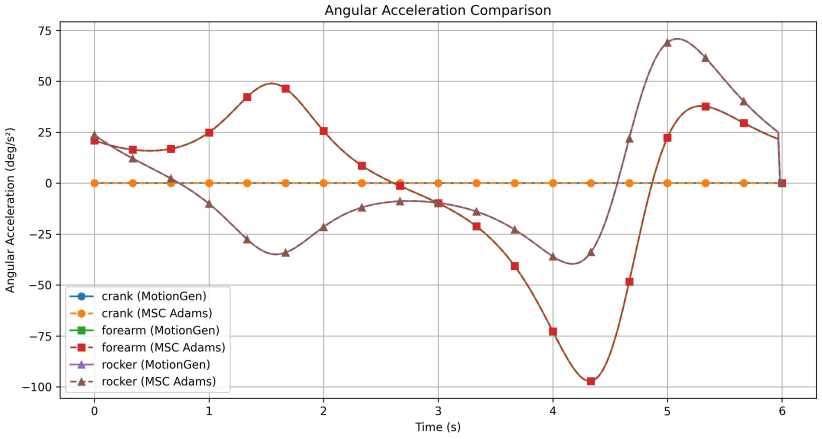


Figure 16: Comparison of angular acceleration versus time for the crank arm, forearm link and the rocker arm

Since the angular displacement, angular velocity and angular acceleration it is safe to assume that the linear velocity and linear acceleration plots also matches accurately for all the links.

8.4 Connection with the Previous ADAMS Plots

The earlier ADAMS post-processor plots were useful for examining individual bodies and components in detail, including angular motion and center-of-mass kinematics. The present comparison plots serve a different but complementary role: they place the major moving links on the same graph so that their relative kinematic trends can be directly compared.

In particular:

- The constant angular velocity and near-zero acceleration of the crank arm match the earlier ADAMS plots of the driving link.
- The oscillatory velocity and acceleration of the forearm and rocker links are consistent with the previously observed non-uniform angular motion of the rehabilitation link and output link.
- The acceleration peaks seen here correspond to the same reversal and rapid-transmission regions identified earlier in the ADAMS-based kinematic analysis.

Therefore, the spreadsheet-based comparison plots do not replace the earlier ADAMS figures; rather, they validate and summarize them in a more compact and comparative manner. Together, both forms of analysis provide a consistent picture of the mechanism's kinematic behavior.

9 Discussion

The combined evidence from the ADAMS model, the physical prototype, and the exported kinematic plots shows that the synthesized mechanism behaves in a manner consistent with a crank–rocker rehabilitation linkage. The prescribed crank input remains nearly uniform, while the forearm and rocker links display bounded oscillatory motion with time-varying velocity and acceleration.

A major strength of the present design is that the oscillatory output is generated passively through the mechanism geometry, without requiring complex input modulation. This makes the mechanism attractive for rehabilitation applications where repeatable guided motion is needed. The physical prototype further supports this conclusion by demonstrating that the mechanism can be fabricated and operated using a simple linkage arrangement.

At the same time, the kinematic analysis indicates that the output motion is not uniform throughout the cycle. In particular, acceleration peaks occur near reversal regions. These are inherent to crank–rocker motion and do not imply failure of the design, but they are practically important because very sharp acceleration changes may lead to discomfort if the mechanism is used directly for therapeutic motion. This suggests that future refinement should focus not only on achieving the correct path, but also on improving motion smoothness near the turning points.

10 Conclusion and Future Work

A kinematic analysis of the synthesized four-bar rehabilitation mechanism was carried out using MSC ADAMS, a fabricated prototype, and exported motion data. The results consistently show that the mechanism functions as a crank–rocker system with a nearly constant-speed input and a bounded oscillatory output suitable for upper-arm rehabilitation motion.

The ADAMS plots confirmed that the input link maintains nearly constant angular velocity, while the forearm and rocker links undergo non-uniform but continuous motion. The linear kinematic results showed that the motion remains essentially planar, validating the modeling assumptions adopted during synthesis. The exported comparison plots further supported these findings and showed close agreement with the detailed ADAMS post-processor results.

The physical prototype demonstrated that the mechanism can be realized practically using laser-cut links and revolute joints, and that the observed motion qualitatively matches the intended behavior. Thus, the mechanism may be considered kinematically successful at the present stage.

Future work will focus on completing the detailed analytical position, velocity, and acceleration analysis, comparing the analytical results with the ADAMS results, and refining the link dimensions or input actuation strategy to reduce acceleration peaks near reversal positions. Such improvements will be important for enhancing motion smoothness and overall suitability for rehabilitation use.

11 Acknowledgments

We would like to express our sincere gratitude to Prof. Vineet Vashista for teaching the concepts of synthesis and analysis of mechanisms and for guiding this project. We also thank the teaching assistants and team members who helped with motion data collection and clarifying questions we had while completing this phase of the project.

12 Links to Data and Motion Generations videos.

- **Optimized Motion Generation Demonstration:** Click here to view video
- **Upper Arm Rehabilitation Prototype Demonstration:** Click here to view video
- **MSC Adams Simulation File** Click here
- **Crank Arm Data:** Click here
- **Rocker Arm Data:** Click here
- **Forearm Link Data:** Click here
- **CAD Model Step File** Click here
- **CAD Drawings** Click here